

personal statement

My journey to Computer Science Education began during my BA in Mathematics & Education, where I discovered how technology could transform student learning. Gaining a proficiency in Python while studying modules like Cryptography (98%), Networks (88%), Modelling (90%) and Applied Statistics (98%) allowed me to develop visualisations as part of my mathematics lessons to help visualise relevant concepts. While teaching in a DEIS (Delivering Equality of Opportunity in Schools) disadvantaged school, I conducted action research on using the classroom response system Plickers to enhance engagement through formative assessment. Being able to inform my own teaching through the formative assessment results, and watching students who had previously struggled become actively engaged in their learning showed me the powerful intersection of technology and education - a realisation that has guided my career path through industry and back to teaching. My motivation for teaching stems from these experiences working with diverse learners. During my teaching placement, I worked with students from various backgrounds, including English as Additional Language learners and members of the Travelling community. This experience taught me the importance of creating an inclusive classroom environment and differentiating instruction to meet individual needs. I received a standing ovation after delivering a three-day workshop titled 'Creative Coding for Maths Makers', where we used Minecraft and Java to demonstrate the practical applications of algorithms and the overlap between mathematical and computing concepts. This led to an unexpected standing ovation from the students, a moment that strengthened my desire to pursue computing education. I chose to deepen my technical expertise through a Higher Diploma in Software Development, achieving First Class Honours with particularly strong performances in Algorithmic & Logical Methods in Java (77%), Internet Programming in HTML/CSS/Javascript (74%), Computer Architecture & Operating Systems in Assembly (72%) and Databases in PostgreSQL (74%). This led to roles as a Software Engineer at FleetOps and TitanHQ, where I gained practical experience across key areas of the Computer Science curriculum. At FleetOps, I worked with cloud computing and the data pipelines involved with Big Data, while at TitanHQ I contributed to cybersecurity solutions in a Windows development environment. This industry experience allows me to bring current, relevant examples to topics like networking, computer architecture, cybersecurity, and software development methodologies. From 2021-2023, I've been putting both my educational and technical skills into practice as a Computer Science Tutor with the Irish charity Tech Steps. In this role, I've developed and delivered programming courses using Python, JavaScript, and Lua using Microsoft MakeCode and Roblox Studio to students aged 10-18 teaching the fundamentals of programming and artificial intelligence, receiving very positive feedback from the students. I've led specialised workshops for Cruinniú na nÓg and Dyspraxia Ireland, adapting my teaching methods to support different learning needs and abilities. My

unique combination of educational training and industry experience has prepared me well for teaching Computing to both GCSE and A-Level according to specifications published by AQA. My BA and Higher Diploma provide a strong foundation for teaching theoretical concepts like algorithms, relational databases and binary representation, while my software development experience ensures I can effectively guide students through practical programming and project work. I'm proficient in several curriculum-supported languages, including Python 3, Java and C#, and have experience using these in educational settings to make complex concepts accessible to all types of learners. Working with diverse student groups has helped me develop teaching strategies that accommodate different learning styles and abilities. When introducing new programming concepts, I've used a mixture of visual aids, practical coding exercises, and real-world examples. The positive feedback I've received when students grasp a challenging concept or successfully complete their first program continues to motivate me. Whether helping students understand fundamental programming principles, explaining computer architecture, or discussing the ethical implications of technology as digital citizens, I aim to create engaging learning experiences that prepare them for both examinations and future careers in computer science. After completing my BA in Mathematics & Education, I registered with the Teaching Council of Ireland as an NQT in Ireland. This does not qualify for Qualified Teaching Status in the UK until I complete an induction period of 200 hours as per the Droichead specification. This is something I haven't completed, as I am much more interested in teaching Computing and ICT than Mathematics but also because only 15% of secondary schools in Ireland offered computer science at Junior Certificate or Leaving Certificate levels in 2023. I am excited about pursuing a PGCE in Computing to garner experience and expertise in teaching Computer Science, gain Qualified Teacher Status, and build a career teaching in schools in England where the subject is far more readily available for students to study. My experience has shown me that effective computing education requires not just technical knowledge but also the ability to make complex concepts accessible and engaging through a well-informed understanding of relevant pedagogical skills and syllabus contexts. I hope for this PGCE to help me coalesce my educational history, industry experience, and teaching practice in the environment of the English school system to further develop the skills and understanding needed to inspire the next generation of computer scientists.